

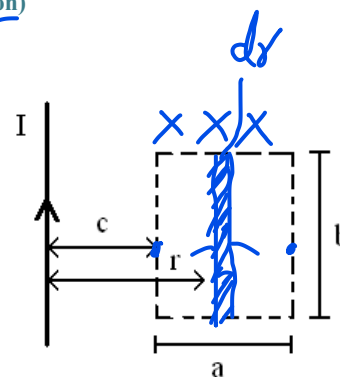
Printable Assignment - Class: PHYS 2100 - Fall 2024 - Malikova Assignment: Ch 23 A (Induction)

Problem 1: An electric current of $I = 0.45$ A flows through a long, straight wire. The wire lies in the plane of a rectangular area that has a pair of sides parallel to the wire. Refer to the diagram. The distance between the wire and the nearest side of the area is $c = 0.037$ m and the area's dimensions are $a = 0.042$ m and $b = 0.054$ m, as shown.

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$$d\phi = b dr \left(\frac{\mu_0 I}{2\pi r} \right)$$

$$\phi = \int_c^{c+a} \frac{b\mu_0 I}{2\pi r} dr = \frac{\mu_0 I b}{2\pi} \int_c^{c+a} \frac{1}{r} dr$$



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Find the magnetic flux, Φ , in $\text{T}\cdot\text{m}^2$, through the area.

Numeric : A numeric value is expected and not an expression.

$\Phi =$ _____

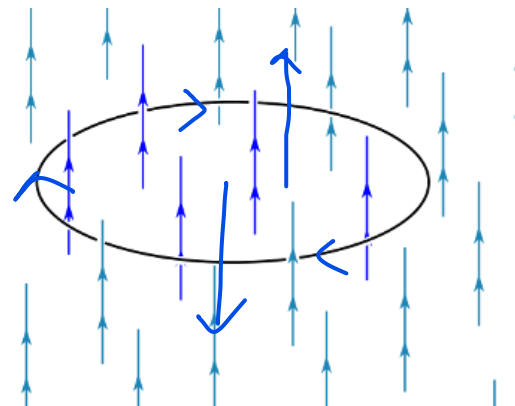
$$= \frac{\mu_0 I b}{2\pi} \ln\left(\frac{c+a}{c}\right)$$

Problem 2: A loop of wire with radius $r = 0.045$ m is placed in a region of uniform magnetic field with magnitude B . As shown in the figure, the field direction is perpendicular to the plane of the loop. The magnitude of the magnetic field changes at a constant rate from $B_1 = 0.65$ T to $B_2 = 7.5$ T in time $\Delta t = 2.5$ s. The resistance of the wire is $R = 11$ Ω .

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$$\mathcal{E} = N \frac{\Delta\Phi_B}{\Delta t} = N \frac{d\Phi_B}{dt} = \frac{N A \Delta B}{\Delta t}$$

$$\mathcal{E} = 1(\pi r^2)(B_2 - B_1) / \Delta t$$



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Part (a) Enter an expression for the magnitude of the magnetic flux through the loop using the symbols in the palette below. Your expression must be valid at all times described in the problem statement.

Expression : $\Phi = B\pi r^2$

Select from the variables below to write your expression. Note that all variables may not be required.

Δt , ϵ_0 , μ_0 , π , B , B_1 , B_2 , e , r , R

Part (b) Enter an expression for the change in the magnitude of the magnetic flux through the loop using the symbols in the palette below.

Expression : $\Delta\Phi = \pi r^2 (B_2 - B_1)$

Select from the variables below to write your expression. Note that all variables may not be required.

Δt , ϵ_0 , μ_0 , π , B , B_1 , B_2 , e , r , R

Part (c) Calculate the numerical value, in tesla squared meters, of the change in the magnitude of the flux.

Numeric : A numeric value is expected and not an expression.

$\Delta\Phi =$ _____ $\text{T} \cdot \text{m}^2$

Part (d) Enter an expression for the magnitude of the induced emf in the loop using the symbols in the palette.

Expression :

$\varepsilon =$ _____

Select from the variables below to write your expression. Note that all variables may not be required.

$\Delta\Phi$, Δt , ε_0 , μ_0 , π , B , B_1 , B_2 , r , R

Part (e) Calculate, in volts, the numerical value of the magnitude of the emf.

Numeric : A numeric value is expected and not an expression.

$\varepsilon =$ _____ V

$$I = \frac{\varepsilon}{R}$$

Part (f) Enter an expression for the magnitude of the current induced in the loop using the symbols in the palette.

Expression :

$I =$ _____

Select from the variables below to write your expression. Note that all variables may not be required.

$\Delta\Phi$, Δt , ε , μ_0 , π , B , B_1 , B_2 , r , R

Part (g) Calculate the numerical value, in amperes, of the magnitude of the induced current.

Numeric : A numeric value is expected and not an expression.

$I =$ _____ A

Part (h) If you are looking down at the loop from above, so that the magnetic field is directed towards you, what is the direction of the induced current?

MultipleChoice :

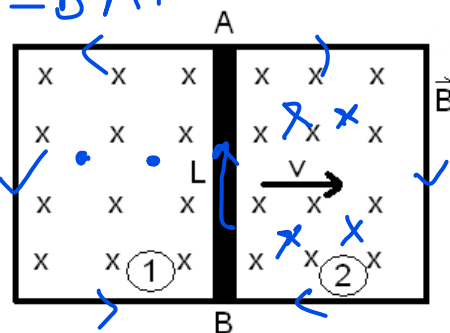
- 1) counterclockwise
- 2) clockwise

$$\frac{\Delta B A}{\Delta t} \quad \varepsilon = \frac{B \Delta A}{\Delta t} = \frac{B \Delta \phi}{\Delta t} = B \frac{\Delta \phi}{\Delta t} = B \frac{\Delta \phi}{\Delta t} \quad \uparrow \phi = B \Delta A$$

Problem 3: A rod AB with length $L = 0.105$ m is lying on a rectangular conducting loop of zero resistance in a magnetic field as shown in the figure. The magnetic field has a constant magnitude $B = 0.65$ T. The rod is moving to the right with speed $v = 0.72$ m/s. The resistance on AB is $R = 130 \Omega$.

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$$\frac{B(Lv\Delta t)}{\Delta t}$$



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Part (a) Express the change of the magnetic flux going through the loop 1, $\Delta\Phi_1$, in terms of B , v , L and Δt .

Expression :

$\Delta\Phi_1 =$ _____

Select from the variables below to write your expression. Note that all variables may not be required.

α , Δt , ε_1 , γ , θ , B , d , g , h , L , m , n , P , R , v

Part (b) Express the magnitude of the average emf induced in loop 1, ε_1 , in terms of B , L and v .

Expression :

$\varepsilon_1 =$ _____

Select from the variables below to write your expression. Note that all variables may not be required.

α , β , Δt , ε_1 , θ , B , d , g , h , L , m , P , R , t , v

Part (c) Calculate the numerical value of ε_1 , in volts.

Numeric : A numeric value is expected and not an expression.

$\varepsilon_1 =$ _____



Part (d) What's the direction of the current induced by ε_1 ?

MultipleChoice :

- 1) Clockwise.
☒ 2) Counterclockwise.

Part (e) Express the change of the magnetic flux going through the loop 2, $\Delta\Phi_2$, in terms of B , v , L and Δt .

Expression :

$\Delta\Phi_2 = -BLv\Delta t$

Select from the variables below to write your expression. Note that all variables may not be required.

α , Δt , ε_1 , γ , θ , B , d , g , h , L , m , n , P , R , v

Part (f) Express the magnitude of the emf induced in loop 2, ε_2 , in terms of B , L and v .

Expression :

$\varepsilon_2 = BLv$

Select from the variables below to write your expression. Note that all variables may not be required.

α , Δt , ε_1 , γ , θ , B , d , g , h , L , m , P , R , t , v

Part (g) Calculate the numerical value of ε_2 , in volts.

Numeric : A numeric value is expected and not an expression.

$\varepsilon_2 = /$

Part (h) What's the direction of the current induced by ε_2 ?

MultipleChoice :

- ☒ 1) Clockwise.
 2) Counterclockwise.

Part (i) Express the current on AB, I , in terms of ε_1 and R .

Expression :

$I = \varepsilon_1 / R$

Select from the variables below to write your expression. Note that all variables may not be required.

α , ε_1 , γ , θ , B , d , g , h , L , m , n , P , R , v , z

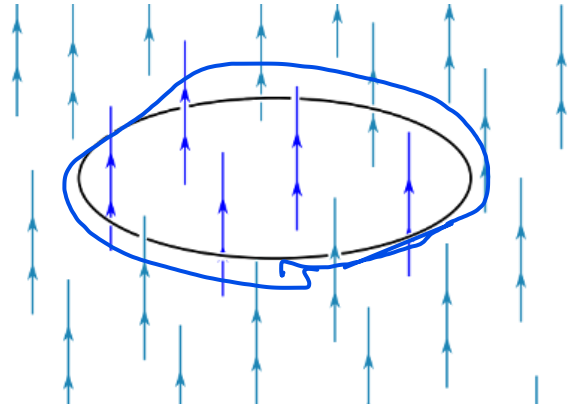
Part (j) Calculate the numerical value of I in A.

Numeric : A numeric value is expected and not an expression.

$I = /$

Problem 4: A loop of wire with radius $r = 0.051$ m is in a magnetic field of magnitude B as shown in the figure. The magnetic field is perpendicular to the plane of the loop. B changes from $B_1 = 0.32$ T to $B_2 = 8.5$ T in $\Delta t = 8.5$ s at a constant rate.

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Part (a) Express the magnetic flux Φ going through a loop of radius r assuming a constant magnetic field B .

Expression :

$\Phi = B\pi r^2$

Select from the variables below to write your expression. Note that all variables may not be required.

$\alpha, \Delta\Phi, \Delta t, \gamma, \pi, \theta, B, B_1, B_2, d, g, h, m, r, t$

Part (b) Express the change in the magnetic flux going through this loop, $\Delta\Phi$, in terms of B_1, B_2 and r .

Expression : $\Delta\Phi = \pi r^2 (B_2 - B_1)$

Select from the variables below to write your expression. Note that all variables may not be required.

$\alpha, \Delta\Phi, \Delta t, \gamma, \pi, \theta, B, B_1, B_2, d, g, h, m, r, t$

Part (c) Calculate the numerical value of $\Delta\Phi$ in $T \cdot m^2$.

Numeric : A numeric value is expected and not an expression.

$\Delta\Phi = \text{ / } T \cdot m^2$

Part (d) Express the magnitude of the average induced electric field, E , induced in the loop in terms of $\Delta\Phi, r$ and Δt .

Expression : $E = \text{ / }$

Select from the variables below to write your expression. Note that all variables may not be required.

$\alpha, \beta, \Delta\Phi, \Delta t, \pi, \theta, B, B_1, B_2, d, g, h, m, n, r$

Part (e) Calculate the numerical value of E in N/C.

Numeric : A numeric value is expected and not an expression.

$E = \text{ / } N/C$

Problem 5: A bicycle can be used to generate electric power by attaching a loop to the axle and letting it rotate in a magnetic field.

Suppose such a device is attached to a bicycle with a 62 cm diameter wheel.

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Faraday's law of induction
 $\mathcal{E} = N \frac{d\Phi_B}{dt}$

$$v = r\omega = 0.31 \times 68 =$$

$$\mathcal{E} = N\omega AB$$

$$16 = 68 \pi (0.31)^2 \times B \Rightarrow B = 0.779 T$$

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Part (a) What is the velocity (in m/s) of the bicycle if the generator's angular velocity is 68 rad/s?

Numeric : A numeric value is expected and not an expression.

$v_1 = 21.08 \text{ m/s}$

$$1 \text{ rev/s} = 2\pi \text{ rad/s}$$

Part (b) What is the maximum emf of the generator when the bicycle moves at 3.5 m/s, noting that it was 16 V under the original conditions?

Numeric : A numeric value is expected and not an expression.

$$\mathcal{E}'_0 = N\omega AB = \left(\frac{3.5}{0.31}\right) \times \pi (0.31)^2 \times 0.779 =$$

Part (c) If the sophisticated generator can vary its own magnetic field, what field strength will it need at 10.00 m/s to produce a 9.00 V maximum emf? Assume the coil has 1.00 by 3.00 cm rectangular coil with 100 turns.

Numeric : A numeric value is expected and not an expression.

$B = 9.3 T$

$$\mathcal{E} = N\omega AB$$

$$9 = 100 \left(\frac{10}{0.31}\right) \times 0.0003 \times B$$

Problem 6: Suppose the emf generated by a coil rotating in a magnetic field is zero at $t = 0$, and increases to its first peak at $t_0 =$

0.075 ms.

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